5.5 PLACENTA AND MEMBRANES

- This is a feto maternal organ. It has two components:
  - Fetal part – develops from the chorionic sac
  - Maternal part – derived from the endometrium

The placenta and the umbilical cord are a transport system for substances between the mother and the fetus.

Structure of placenta: It is a flat, round mass, about 15 to 20 cm in diameter, 2.5-3 cm thickness, 15-20 lobes, weighs 1/6th of baby's weight or 500 – 600gms at birth. It has two surfaces maternal surface and fetal Surface

Maternal surface

It is irregular, and divided into convex areas (cotyledons)

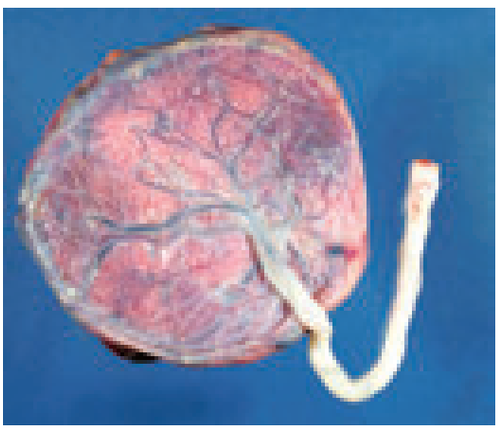
- Cotyledons –about 15 to 20 slightly bulging villous areas. Their surface is covered by shreds of decidua basalis from the uterine wall.
- After birth, the placenta is always carefully inspected for missing cotyledons. Cotyledons remaining attached to the uterine wall after birth may cause severe bleeding.



Maternal surface of placenta

Fetal surface:

- This side is smooth and shiny. It is covered by amnion.
- The umbilical cord is attached close to the center of the placenta.
- The umbilical vessels radiate from the umbilical cord.
- They branch on the fetal surface to form chorionic vessels.
- They enter the chorionic villi to form arteriocapillary-venous system.



Fetal surface of placenta

Placental Membranes

- The placental membrane separates maternal blood from fetal blood. The fetal part of the placenta is known as the chorion. The maternal component of the placenta is known as the decidua basalis.

Fetal membranes: It consists of two layers.

- **Chorion:** It is the outer layer of fetal membranes. it is thick friable and shaggy.
- **Amnion:** It is the inner layer of fetal membrane. It is smooth, shiny , and transparent
- Placental Circulation Fetal – from Umbilical Arteries to chorionic plate to branches to stem villi to capillaries in terminal villi and return via umbilical vein.
- Placental circulation Maternal – Free-flowing with Spiral arteries open into intervillous space and bath the villi with 150 ml of maternal

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